Project Validation Report: Real-Time Object Detection in Urban Traffic Scenes

Date: July 10, 2026
Auditor: Independent Research Auditor (AI System Instance)
Status: Provisional (Training in Progress)
Target Project: object-Detection

SECTION 1: PROJECT SUMMARY

Project Objective

The objective of this project is to develop and validate a robust, real-time urban traffic object detection system suitable for autonomous driving and smart-city applications. It aims to deliver high-accuracy detections across multiple categories of road users and vehicles, operating efficiently under varying lighting conditions and dense city environments.

Research Problem

Urban traffic scenes present severe challenges for standard computer vision models: 1. **Vulnerability & Scales:** Small, vulnerable road users (e.g., pedestrians, riders, cyclists) are easily obscured and exhibit high visual variance. 2. **Regional Variance:** Standard driving datasets (e.g., COCO, BDD100K) lack common regional transport vehicles such as three-wheeled auto-rickshaws, limiting their deployment in developing economies like India. 3. **Class Imbalance:** Extreme class distribution differences (e.g., a high volume of cars vs. very few riders) lead to model bias and high false-negative rates for minority classes.

Novel Contribution

The repository addresses these problems through a unified workflow consisting of: - **Dataset Integration & Fusion:** Merging the Berkeley BDD100K base dataset with the India Driving Dataset (IDD) and a custom Auto-rickshaw dataset, mapping them into a unified 10-class label space. - **Class Balancing:** Automated oversampling of underrepresented classes (specifically the `rider` class). - **Hard-Example Mining:** An automated mining pipeline that isolates low-confidence inferences and false negatives from validation runs, allowing for targeted model fine-tuning. - **FastAPI Visualizer:** An interactive, glassmorphic live stream dashboard for real-time visualization and user control of detection thresholds.

Detection Pipeline

1. Input frame from webcam, uploaded video, or stream.
2. Frame preprocessed (resize, normalize) and passed to the custom YOLO model wrapper UrbanDetector.
3. The model outputs class bounding boxes and confidence scores.
4. Non-Maximum Suppression (NMS) removes duplicate detections.
5. Detections are overlaid on the video stream and rendered inside cv2 GUI or streamed to the FastAPI frontend.

Dataset Pipeline

1. Raw BDD100K labels converted to YOLO format via bdd_to_yolo_prod.py.
2. Converted label-image mappings verified using verify_dataset.py.
3. XML auto-rickshaw annotations converted via convert_auto_to_yolo.py.
4. IDD labels remapped and merged via merge_idd.py.
5. Auto-rickshaw labels merged via merge_auto_yolo.py.
6. Minority class oversampling applied via oversample_rider.py.

Training Pipeline

A multi-stage transfer learning strategy is employed using YOLOv8l: 1. **Stage 1:** Base model initialized on COCO weights and pre-trained on BDD100K base classes for 30 epochs. 2. **Stage 2:** Fine-tuning continued on BDD100K for 70 additional epochs (total 100). 3. **Stage 3:** Model trained on mined hard-examples (BDD without train class) with stronger augmentations. 4. **Stage 4:** Final tuning on BDD Balanced (merged with IDD and Auto-rickshaw, plus rider oversampling) for 50 epochs.

Evaluation Pipeline

Runs validation on target datasets after training, computing standard metrics (Precision, Recall, mAP@50, mAP@50-95, box/cls/dfl losses), and automatically plotting the Confusion Matrix, PR Curves, and Per-Class AP lists.

Deployment & Dashboard Pipeline

FastAPI web app (`src/dashboard/app.py`) starts a background thread running CameraManager. The manager reads video frames, feeds them to UrbanDetector, caches predictions and performance stats, encodes frames as JPEG, and streams them to the glassmorphic HTML/JS interface via a StreamingResponse.

SECTION 2: DATASET INFORMATION

Dataset Details

- **Dataset Names:**
- BDD100K (Berkeley DeepDrive) Subset
- IDD (India Driving Dataset) Subset
- Datacluster Labs Auto-Rickshaw Dataset
- **Dataset Versions:** Standard release variants remapped to YOLO formats.
- **Number of Datasets:** 3 distinct sources.
- **Training Images:**
- Base BDD subset (`bdd_yolo`): **1,153** images
- Merged Balanced subset (`bdd_balanced`): **34,536** images
- **Validation Images:**
- Base BDD subset (`bdd_yolo`): **9,999** images
- Merged Balanced subset (`bdd_balanced`): **14,150** images
- **Test Images:** `NOT FOUND` (The local workspace does not contain test splits).
- **Number of Classes:** 10 classes in both active schemas.
- **Class Names:**
- **BDD100K Base:** `pedestrian`, `rider`, `car`, `truck`, `bus`, `train`, `motorcycle`, `bicycle`, `traffic light`, `traffic sign`
- **BDD Balanced:** `pedestrian`, `rider`, `car`, `truck`, `bus`, `motorcycle`, `bicycle`, `traffic light`, `traffic sign`, `auto`

Dataset Merge Workflow

- **IDD Integration:** Remapped 13 original IDD class IDs to BDD equivalents using merge_idd.py. Remapped index `1` (autorickshaw) -> `9` (auto), caravan and trailer to `truck`, and person to `pedestrian`.
- **Auto-rickshaw Integration:** Converted Pascal VOC XML bounding boxes (`autorickshaw` / `auto`) to normalized YOLO coordinates with class ID `9` via convert_auto_to_yolo.py. Integrated them using merge_auto_yolo.py with an 80/20 train/val split.
- **Oversampling Strategy:** Duplicated training samples containing the rare `rider` class (class ID 1) by a factor of 2 via oversample_rider.py to reduce class imbalance.

Folder Structure

```
object-Detection/data1/
|-- bdd100k/                        # Raw BDD100K subset images & labels
|-- IDDDetectionsYOLODataset/       # Raw IDD subset files (YOLO format)
|-- auto/                           # Auto-rickshaw images
|-- Annotations/                    # Auto-rickshaw XML annotations
|-- auto_yolo/                      # Converted auto-rickshaw YOLO files
|-- bdd_yolo/                       # BDD100K base dataset (YOLO format)
|-- bdd_balanced/                   # Unified final merged dataset
|   |-- images/
|   |   |-- train                   # 34,536 images (.jpg + .npy cache)
|   |   `-- val                     # 14,150 images (.jpg + .npy cache)
|   `-- labels/
```

```
|       |-- train                  # 34,536 text labels
|       `-- val                    # 14,150 text labels
|-- hard_examples/                 # Missed and low-confidence mined frames
```

Dataset YAML Configurations

1. bdd100k.yaml

```
path: ./bdd_yolo
train: images/train
val: images/val

nc: 10

names:
  0: pedestrian
  1: rider
  2: car
  3: truck
  4: bus
  5: train
  6: motorcycle
  7: bicycle
  8: traffic light
  9: traffic sign
```

2. bdd_balanced.yaml

```
path: c:/Repo/object-Detection/data1/bdd_balanced

train: images/train
val: images/val

nc: 10

names:
  0: pedestrian
  1: rider
  2: car
  3: truck
  4: bus
  5: motorcycle
  6: bicycle
  7: traffic light
  8: traffic sign
  9: auto
```

SECTION 3: MODEL INVENTORY

A comprehensive search of the repository reveals the following models:

1. YOLOv8l (object_stage1)

- **Exists:** YES
- **Trained:** YES (Completed)
- **Training Status:** Completed
- **Weight Path:** best.pt

- **Dataset Used:** `data1/bdd100k.yaml`
- **Epochs:** 30
- **Batch Size:** 8
- **Image Size:** 832
- **Optimizer:** auto
- **Device:** GPU (device 0)
- **Date Trained:** Feb 28, 2026
- **Checkpoints:** Best & Last

2. YOLOv8l (object_stage2)

- **Exists:** YES
- **Trained:** YES (Completed)
- **Training Status:** Completed
- **Weight Path:** best.pt
- **Dataset Used:** `data1/bdd100k.yaml` (from Stage 1 weights)
- **Epochs:** 70
- **Batch Size:** 8
- **Image Size:** 832
- **Optimizer:** auto
- **Device:** GPU (device 0)
- **Date Trained:** Feb 28, 2026
- **Checkpoints:** Best & Last

3. YOLOv8l (object_stage3_strong2)

- **Exists:** YES
- **Trained:** YES (Completed)
- **Training Status:** Completed
- **Weight Path:** best.pt
- **Dataset Used:** `data1/bdd_no_train.yaml` (from Stage 2 weights)
- **Epochs:** 50
- **Batch Size:** 6
- **Image Size:** 960
- **Optimizer:** auto
- **Device:** GPU (device 0)
- **Date Trained:** Mar 02, 2026
- **Checkpoints:** Best & Last

4. YOLOv8l (object_stage4_riderfix4)

- **Exists:** YES
- **Trained:** YES (Completed)
- **Training Status:** Completed
- **Weight Path:** best.pt
- **Dataset Used:** `data1/bdd_balanced.yaml` (from Stage 3 weights)
- **Epochs:** 50
- **Batch Size:** 6
- **Image Size:** 960
- **Optimizer:** auto
- **Device:** GPU (device 0)
- **Date Trained:** Mar 05, 2026
- **Checkpoints:** Best & Last

5. YOLOv8l (train_fast2)

- **Exists:** YES
- **Trained:** YES (Completed)

- **Training Status:** Completed
- **Weight Path:** best.pt
- **Dataset Used:** `data1/bdd_balanced.yaml` (from Stage 3 weights)
- **Epochs:** 30
- **Batch Size:** 32
- **Image Size:** 512
- **Optimizer:** auto
- **Device:** GPU (device 0)
- **Date Trained:** Mar 10, 2026
- **Checkpoints:** Best & Last

6. YOLOv8l (train_auto_v1)

- **Exists:** YES
- **Trained:** YES (Completed)
- **Training Status:** Completed
- **Weight Path:** best.pt
- **Dataset Used:** `data1/bdd_balanced.yaml`
- **Epochs:** 50
- **Batch Size:** 4
- **Image Size:** 640
- **Optimizer:** auto
- **Device:** CPU (None)
- **Date Trained:** Mar 06, 2026
- **Checkpoints:** Best & Last

7. YOLOv8n (train3)

- **Exists:** YES
- **Trained:** YES (Completed)
- **Training Status:** Completed
- **Weight Path:** best.pt
- **Dataset Used:** `data/bdd100k.yaml`
- **Epochs:** 10
- **Batch Size:** 16
- **Image Size:** 640
- **Optimizer:** auto
- **Device:** CPU
- **Date Trained:** Feb 15, 2026
- **Checkpoints:** Best & Last

8. YOLOv5s (validation_yolov5s)

- **Exists:** YES
- **Trained:** YES (Completed)
- **Training Status:** Completed (Sanity check)
- **Weight Path:** best.pt
- **Dataset Used:** `data1/bdd_balanced.yaml`
- **Epochs:** 1
- **Batch Size:** 16
- **Image Size:** 640
- **Optimizer:** SGD
- **Device:** GPU (device 0)
- **Date Trained:** Jul 10, 2026
- **Checkpoints:** Best & Last

9. YOLOv5s (train_yolov5s_v1)

- **Exists:** YES
- **Trained:** NO (Training in progress)
- **Training Status:** TRAINING IN PROGRESS – FINAL RESULT PENDING
- **Weight Path:** None (active checkpoint written to weights)
- **Dataset Used:** `data1/bdd_balanced.yaml`
- **Epochs:** 50 (Target)
- **Batch Size:** 16
- **Image Size:** 640
- **Optimizer:** SGD
- **Device:** GPU (device 0)
- **Date Trained:** Active (Started Jul 10, 2026 15:35)
- **Checkpoints:** Active (`last.pt` / `best.pt` changing dynamically)

Other Models Search

- **YOLOv5m:** `NOT IMPLEMENTED` (No configs, weights, or logs exist in the repository).
- **YOLOv7x:** `NOT IMPLEMENTED` (No configs, weights, or logs exist in the repository).
- **YOLOv8s:** `NOT IMPLEMENTED` (The directory `train_yolov8s` exists but contains no logs or weights).
- **YOLOv8m:** `NOT IMPLEMENTED` .

SECTION 4: TRAINING RESULTS

The metrics below are extracted from local training run records (`results.csv` , `opt.yaml` , `args.yaml`) in `runs/detect` and `experiments/yolov5` :

Run Identifier	Model	Epochs	Batch	Imgsz	Precision	Recall	mAP@50	mAP@50-95	Train/Val Loss	Time (s)	Checkpoi
object_stage1	YOLOv8l	30	8	832	0.6443	0.4012	0.4106	0.2256	1.169 / 1.373	4,078	`best.pt` `last.pt`
object_stage2	YOLOv8l	70	8	832	0.5058	0.3834	0.3901	0.2121	0.858 / 1.443	9,219	`best.pt` `last.pt`
object_stage3_strong2	YOLOv8l	50	6	960	0.6116	0.4538	0.4732	0.2552	0.988 / 1.493	8,354	`best.pt` `last.pt`
object_stage4_riderfix4	YOLOv8l	50	6	960	0.5998	0.4463	0.4618	0.2572	0.655 / 1.483	8,672	`best.pt` `last.pt`
train_fast2	YOLOv8l	30	32	512	0.7147	0.5065	0.5524	0.3480	0.973 / 1.227	48,078	`best.pt` `last.pt`
train_auto_v1	YOLOv8l	50	4	640	0.5554	0.4742	0.4828	0.2952	1.049 / 1.365	5,205	`best.pt` `last.pt`
train6	YOLOv8l	50	8	832	0.6526	0.3983	0.4086	0.2244	1.045 / 1.404	6,856	`best.pt` `last.pt`
train3	YOLOv8n	10	16	640	0.4049	0.0019	0.2034	0.1220	0.470 / 1.862	144,025	`best.pt` `last.pt`
validation_yolov5s	YOLOv5s	1	16	640	0.5072	0.2769	0.2726	0.1494	0.086 / 0.071	Fast	`best.pt` `last.pt`
train_yolov5s_v1	YOLOv5s	50	16	640	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDIN

> [!NOTE] **Active YOLOv5s Training Status:** The training run `train_yolov5s_v1` is currently active. The last logged metrics at **Epoch 10** are: - **Precision:** 0.65861 - **Recall:** 0.42751 - **mAP@50:** 0.47126 - **mAP@50-95:** 0.27244 - **Train Box/Obj/Cls Loss:** 0.067867 / 0.050062 / 0.024563 - **Val Box/Obj/Cls Loss:** 0.062933 / 0.071744 / 0.014115 - Final metrics and checkpoints are: TRAINING IN PROGRESS – FINAL RESULT PENDING

Visual Artifacts

- **Confusion Matrix:** Automatically generated for completed models and stored in the respective run folder (e.g. `runs/detect/object_stage4_riderfix4/confusion_matrix.png`).
- **PR Curves:** Automatically generated and saved in run folders (e.g. `validation_yolov5s/PR_curve.png`).
- **Per-Class AP:** Calculated during validation steps and saved in run logs.

SECTION 5: PROJECT FEATURES

Below is the verification status of core project pipeline components:

Feature / Pipeline	Status	Evidence	Notes
Dataset Preprocessing	**IMPLEMENTED**	`src/bdd_to_yolo.py`	Basic conversion scripts
Dataset Verification	**IMPLEMENTED**	verify_dataset.py	Checks folder alignment
Dataset Merging	**IMPLEMENTED**	merge_idd.py, merge_auto_yolo.py	Remaps and fuses IDD/Auto
Oversampling	**IMPLEMENTED**	oversample_rider.py	Rider class duplicator
Auto-Rickshaw Integration	**IMPLEMENTED**	convert_auto_to_yolo.py	XML parser to YOLO format
IDD Integration	**IMPLEMENTED**	merge_idd.py	Class mapping script
BDD100K Conversion	**IMPLEMENTED**	`src/bdd_to_yolo_prod.py`	Image copy workflow
YOLOv5 Support	**PARTIAL**	`scripts/train_yolov5s.py`	Submodule train code, active YOLOv5s training
YOLOv7 Support	**NOT IMPLEMENTED**	None	No YOLOv7 model code or scripts found
YOLOv8 Support	**IMPLEMENTED**	`src/detector.py`, `src/predict_webcam.py`	Native Ultralytics implementation
Training Pipeline	**IMPLEMENTED**	`scripts/train_orchestrator.py`	Orchestration scripts for multi-stage runs
Inference Pipeline	**IMPLEMENTED**	main.py, `src/detector.py`	OpenCV frame rendering
Evaluation Pipeline	**IMPLEMENTED**	YOLO `val` commands	Generates logs, CSV metrics, and curves
Benchmark Pipeline	**NOT IMPLEMENTED**	None	No speed/accuracy comparison suite found
FastAPI Dashboard	**IMPLEMENTED**	`src/dashboard/app.py`	Live preview web API
Camera Manager	**IMPLEMENTED**	`src/dashboard/camera_manager.py`	Threaded frame streamer
Detector API	**IMPLEMENTED**	`src/detector.py`	Wrapper class Around YOLO
Web Interface	**IMPLEMENTED**	`src/dashboard/static/index.html`	Glassmorphic frontend
Model Comparison	**NOT IMPLEMENTED**	None	No qualitative / quantitative comparison scripts
Qualitative Comparison	**NOT IMPLEMENTED**	None	No side-by-side verification figures
Deployment Configuration	**NOT IMPLEMENTED**	None	No Docker or container setups

SECTION 6: PAPER REPRODUCIBILITY

This section checks whether claims in the paper's outline can be reproduced using the current repository state:

Paper Claim	Repository Evidence	Status	Notes
Dataset Configuration	bdd_balanced.yaml	✓ MATCH	Contains 10 final classes matching target schema
Class Mapping	merge_idd.py	✓ MATCH	Implements the IDD-to-BDD mapping correctly
YOLOv5s	`train_yolov5s.py`, `train_yolov5s_v1/`	⚠ PARTIAL	Training is actively running; results pending
YOLOv5m	None	✗ NOT FOUND	No YOLOv5m configs, weights, or references found
YOLOv7x	None	✗ NOT FOUND	No YOLOv7 architecture support found in repo
YOLOv8n	`runs/detect/train3` weights	✓ MATCH	Trained successfully for 10 epochs on CPU
YOLOv8s	`runs/detect/train_yolov8s` (empty)	✗ NOT FOUND	Folder exists but contains no training runs
YOLOv8m	None	✗ NOT FOUND	No YOLOv8m configs, weights, or references found
YOLOv8l	`object_stage4_riderfix4/weights/best.pt`	✓ MATCH	Fully trained through all 4 stages
Training Pipeline	`scripts/train_orchestrator.py`	✓ MATCH	Orchestration workflow runs Stage 1-4 training
Evaluation Metrics	`results.csv` files	✓ MATCH	Standard YOLO verification CSVs present
Benchmark Metrics	None	✗ NOT FOUND	No comparison or benchmarking scripts
Inference Scripts	main.py	✓ MATCH	Standard webcam/video inference wrapper
FastAPI Dashboard	`src/dashboard/app.py`	✓ MATCH	Glassmorphic active dashboard interface
Deployment Setup	None	✗ NOT FOUND	No production cloud or container config files
Qualitative Fig	None	✗ NOT FOUND	No visual detection comparisons between model versions
Confusion Matrix	`confusion_matrix.png` files	✓ MATCH	Automatically output by Ultralytics val runs
PR Curve	`PR_curve.png` files	✓ MATCH	Automatically output by Ultralytics val runs
Per-Class AP	Val logs	✓ MATCH	Per-class AP available in log archives

Explanations of Mismatches

1. **YOLOv5m, YOLOv7x, YOLOv8m:** Not implemented. No training code, weights, or configuration parameters are defined for these architectures in the repository.
2. **YOLOv8s:** A directory was created but no training was ever executed (empty folders, no checkpoints or logs).
3. **YOLOv5s:** Training is actively running. Final validation metrics are pending.
4. **Benchmark & Qualitative Fig:** No comparison script comparing cross-architecture inference speeds (FPS) or side-by-side detection images exists.

SECTION 7: METRIC COMPARISON

> [!WARNING] **Research Paper Draft Missing:** The conference paper draft was NOT FOUND in the workspace. No draft metrics tables or baseline figures are available in the project directories to compile a direct validation comparison.
> Furthermore, the YOLOv5s training run is currently active, which blocks final metric logging for the comparison models.

Metric	Paper Value	Repository Value	Difference	% Var	Status
YOLOv5s mAP@50	Unknown	PENDING	N/A	N/A	TRAINING IN PROGRESS – FINAL RESULT PENDING
YOLOv5s Precision	Unknown	PENDING	N/A	N/A	TRAINING IN PROGRESS – FINAL RESULT PENDING
YOLOv5s Recall	Unknown	PENDING	N/A	N/A	TRAINING IN PROGRESS – FINAL RESULT PENDING
YOLOv8l mAP@50	Unknown	0.4618 (stage4)	N/A	N/A	Final Baseline Logged (0.46175)
YOLOv8l Precision	Unknown	0.5998 (stage4)	N/A	N/A	Final Baseline Logged (0.59976)
YOLOv8l Recall	Unknown	0.4463 (stage4)	N/A	N/A	Final Baseline Logged (0.44633)

SECTION 8: QUALITATIVE VALIDATION

Reviewer Requirement

The reviewer requested a qualitative detection comparison displaying side-by-side inference outputs of the: 1. **Original Image** (Ground Truth / Unannotated) 2. **YOLOv5s Detections** 3. **YOLOv8l Detections**

Implementation Status

This comparison has **NOT BEEN IMPLEMENTED**.

Remaining Tasks to Resolve Reviewer Requirements

1. **Inference Script:** A Python comparison script (e.g. `scripts/generate_qualitative_figs.py`) needs to be created. This script should load `yolov5s` and `yolov8l` weights, run inference on the same subset of validation images, and save combined comparison figures.
2. **Comparison Image Generation:** Export images showing side-by-side bounding boxes for road scenes containing minority classes (`rider`, `auto`) during day, night, and heavy traffic conditions.
3. **Documentation:** Insert these comparison figures directly into the results folder and include a description in the paper.

SECTION 9: REPOSITORY HEALTH

An evaluation of the repository health is detailed below:

Audit Category	Score (1-10)	Auditor Rationale & Explanations
Folder Structure	**8 / 10**	Well-separated datasets, runs, src, and script dirs. However, old duplicate scripts in the root directory (`oversample_rider.py`, `downsample_bdd.py`, `remove_train_class.py`) conflict with `src/` files and clutter the root.
Architecture	**8.5 / 10**	Clear modular separation of pipeline stages: conversion scripts, training orchestrator, and FastAPI dashboard camera threads are well isolated.

Audit Category	Score (1-10)	Auditor Rationale & Explanations
Documentation	**7.5 / 10**	Clean setup guides (`README.md`, `IMPLEMENTATION_GUIDE.md`) exist, but there is no paper draft or explanation of training run histories/configs.
Code Quality	**8.5 / 10**	Uses clean standard libraries, structured arguments (`argparse`), progress bars (`tqdm`), and correct thread safety locks inside `camera_manager.py`.
Maintainability	**8 / 10**	Clear script naming, but duplicate code in the root directory risks configuration drift and increases maintenance cost.
Scalability	**8 / 10**	Pipeline handles large directories easily. Could be improved by moving hardcoded folders in scripts to central environment configuration files.
Reproducibility	**7 / 10**	The YOLOv5 orchestrator records `environment.json` and `experiment.yaml`. However, custom YOLOv8 stages lack documented seeds and parameters.
Deployment Readiness	**6.5 / 10**	Local FastAPI server works cleanly. There are no production configuration setups, environment variables, or Docker containerizations.
Research Repro.	**6 / 10**	Baseline paper metrics tables are missing from the repo, preventing direct verification of model performance claims.

SECTION 10: MISSING ITEMS

Below is a checklist of tasks still missing from the repository: 1. **YOLOv5m Model:** No training scripts, configurations, or weights exist. 2. **YOLOv7x Model:** Completely missing from the repository files. 3. **YOLOv8s Model:** Training folder `train_yolov8s` is empty; model was never trained. 4. **YOLOv8m Model:** Completely missing from the repository files. 5. **Qualitative Figure Generator:** Visual side-by-side comparison figures comparing Original vs. YOLOv5s vs. YOLOv8l are not generated. 6. **Benchmark Pipeline:** A tool to compare inference latency, speed (FPS), and mAP across trained versions is missing. 7. **Production Deployment Configs:** Missing Docker container configurations (`Dockerfile` / `docker-compose.yml`) or production server settings. 8. **Conference Paper Draft:** The actual LaTeX or Word draft of the conference paper is missing.

SECTION 11: ITEMS PENDING COMPLETION

The following items are blocked and cannot be finalized because the YOLOv5s training run (`train_yolov5s_v1`) is actively running: 1. **Current Epoch:** Training is at **Epoch 10 / 50**. 2. **Final mAP:** Final mAP@50 and mAP@50-95 values are pending. 3. **Final Precision:** Final precision score is pending. 4. **Final Recall:** Final recall score is pending. 5. **Final Checkpoint:** The final optimized weight file (`best.pt`) is pending. 6. **Final validation report logs:** Post-training validation evaluations are pending. 7. **PR Curve & Confusion Matrix plots:** Final curve charts for YOLOv5s are pending. 8. **Qualitative Figures:** Generation of YOLOv5s detections for the reviewer comparison requires the final checkpoint.

SECTION 12: SUPERVISOR SUMMARY

To: Research Supervisor
From: Independent Research Auditor
Date: July 10, 2026
Subject: Technical Audit of YOLO Urban Object Detection Repository

Project Status Overview

This audit evaluated the `object-Detection` repository to compare current implementation states against the conference research claims. The pipeline is **PARTIALLY COMPLETED**. Core modules for dataset preprocessing, 1:1 integrity verification, custom IDD/Auto remapping, and class balancing have been successfully implemented.

However, a critical YOLOv5s training run is **actively running** on the system, blocking final metric validation. Furthermore, several models (YOLOv5m, YOLOv7x, YOLOv8s, YOLOv8m) and reviewer requirements (qualitative comparison figures) remain **NOT IMPLEMENTED**.

Status Breakdown

1. COMPLETED

- **Dataset Preprocessing:** Raw BDD100K JSON conversion scripts are functional.
- **Dataset Merging:** remaps and integrates IDD and Auto-rickshaw datasets into BDD.
- **Class Balancing:** Oversampling minority classes (duplicating `rider` images) is completed.
- **Hard-Example Mining:** Isolated missed and low-confidence detections from validation runs.
- **YOLOv8l & YOLOv8n Models:** Multi-stage training (Stage 1 to 4) of YOLOv8l is fully completed.
- **FastAPI Dashboard:** Live streaming dashboard with conf slider controls, uploader, and threaded CameraManager is fully functional.

2. IN PROGRESS

- **YOLOv5s Training:** The training orchestrator is actively running `train_yolov5s_v1` on GPU device 0 (Epoch 10 of 50 completed).
- **YOLOv5s Metrics & Plots:** results.csv log updating in real-time. Final metrics are pending.

3. NOT IMPLEMENTED

- **Reviewer Qualitative Figures:** Side-by-side comparison images of Original vs. YOLOv5s vs. YOLOv8l have not been generated.
- **Benchmarking Suite:** A consolidated benchmarking tool for measuring FPS and speed differences is missing.
- **Other Models:** YOLOv5m, YOLOv7x, YOLOv8s, YOLOv8m training has not been executed.
- **Deployment configs:** Docker container configs are missing.
- **Paper Draft Document:** The draft of the conference paper is not stored in the repository.

SECTION 13: CURRENT STATUS

Based on the actual state of the repository, the project is classified as:

PARTIALLY READY (Provisional Assessment)

Rationale

The repository contains functional dataset conversion and fusion pipelines, completed multi-stage training runs for the YOLOv8l architecture, and a working FastAPI dashboard. However, since the YOLOv5s training job is actively running, final metrics cannot yet be logged. The reviewer's qualitative comparison request and benchmarking suite remain unimplemented.

This assessment is PROVISIONAL and subject to change after the active training finishes and the reviewer requirements are resolved.